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BIF-501 Bioinformatics II



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Word Limit: Not given

Solution:

Impact of Machine Learning on Bioinformatics:

Machine learning in bioinformatics is the application of machine learning algorithms to bioinformatics, including genomics, proteomics, microarrays, systems biology, evolution, and text mining.

Prior to the emergence of machine learning, bioinformatics algorithms had to be programmed by hand; for problems such as protein structure prediction, this proved difficult.Machine learning techniques, such as deep learning can learn features of data sets, instead of requiring the programmer to define them individually. The algorithm can further learn how to combine low-level features into more abstract features, and so on. This multi-layered approach allows such systems to make sophisticated predictions when appropriately trained.

Five Major impacts of Machine Learning on Bioinformatics are

1.Facilitating gene editing experiments

Gene editing refers to manipulations on an organism's genetic composition by deleting, inserting, and replacing a part of its DNA sequence. This process typically relies on the CRISPR technique, which is rather effective. But there is still much improvement to be desired in the area of selecting the right DNA sequence for manipulation, and this is where ML can help. Using machine learning for bioinformatics, researchers can enhance the design of gene editing experiments and predict their outcomes.

2. Identifying protein structure

Proteomics is a study of proteins, their interactions, composition, and their role in the human body. This field involves heavy biological datasets and is computationally expensive. Therefore, technologies like machine learning in bioinformatics are essential here.

One of the most successful applications in this field is using convolutional neural networks to position proteins' amino acids into three classes — sheet, helix, and coil. Neural networks can achieve an accuracy of 84% with the theoretical limit being 88%–90%.

3. Spotting genes associated with diseases

Researchers increasingly use machine learning in bioinformatics to identify genes that are likely to be involved in particular diseases. This is achieved by analyzing gene expression microarrays and RNA sequencing.

Particularly, gene identification gains traction in cancer-related studies to identify genes that are likely to contribute to cancer, as well as classify tumors by analyzing them on a molecular level.

4. Traversing the knowledge base in search of meaningful patterns

Advanced sequencing technology doubles genomic databases each 2.5 years, and researchers are looking for a way to extract useful insights from this accumulated knowledge. Machine learning in bioinformatics can sift through biomedical publications and reports to identify different genes and proteins and search for their functionality. It can also aid in annotating protein databases and complement them with the information it retrieves from the literature.

5.Repurposing drugs

Drug repurposing, or reprofiling, is a technique scientists use to discover new applications of existing drugs that they were not intended for. Researchers adopt AI in bioinformatics to perform drug analysis on relevant databases, such as BindingDB and DrugBank.

Regards:

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